

— AUTONOMOUS QUALITY ENGINEERING · BUILT TO SCALE ACROSS LBG

Meet  **Helios.**

An autonomous Quality Engineering AI Agent.

Helios — for the sun and illumination: clarity and visibility in quality automation.

From a single Jira ticket to **generated test scenarios, triggered pipelines** and **trusted, auditable evidence written straight back to the ticket** — with minimal human effort. A proof of concept built to drive the MVP inside the Economic Crime Prevention Platform.

● Google ADK + Gemini on Vertex AI

● GKE

● Harness CI/CD

● Jira / Xray

● GitHub · BigQuery · GCS · Neo4j

SCOPE

PoC → MVP blueprint

REACH

ECP: 4 labs · ~40 teams

ADOPTION

Built once · reused as a template

FOUNDATION FOR

Group-wide rollout

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Testing can't keep pace with how fast ECP changes.

ECP products evolve rapidly across complex data flows, integrations and user journeys. Traditional testing leans heavily on manual effort, deep product knowledge and early, clear requirements — slowing feedback and increasing delivery risk.

4

labs across the platform

~40

teams solving the same
testing problem
independently

100%

manual test design today
before a change can ship
with confidence

Where teams lose time today

- ! Heavy reliance on manual test design and deep product knowledge
- ! Slow feedback on whether a change is safe to release
- ! Duplicated effort — every team builds its own approach
- ! Inconsistent quality and uneven test evidence
- ! Coverage depends on requirements being clear and early

Give **Helios** a product change — a Jira story, acceptance criteria, an API or config update — and it decides *what* to test and *why*, runs meaningful automated tests, and produces clear evidence stakeholders can trust. Built once and offered as a standard template: a **single QE can adopt it in one sprint — no extra engineer needed**.



Generate

Reads acceptance criteria and authors Cucumber/BDD scenarios plus Jira *Test* subtasks — opening a GitHub PR with the new features.



Execute

Triggers the Harness testing pipelines automatically when a ticket moves to *Testing*, running deterministic + BDD suites against the live pipeline.



Evidence

Writes pass/fail results back onto the ticket — updating Xray or regular subtasks — and triages failures into follow-up PRs. Fully auditable.

SOURCES OF TRUTH

 Jira / Xray



GKE · HELIOS

 ADK + Gemini orchestrator



EXECUTE & GATE

 Harness pipelines

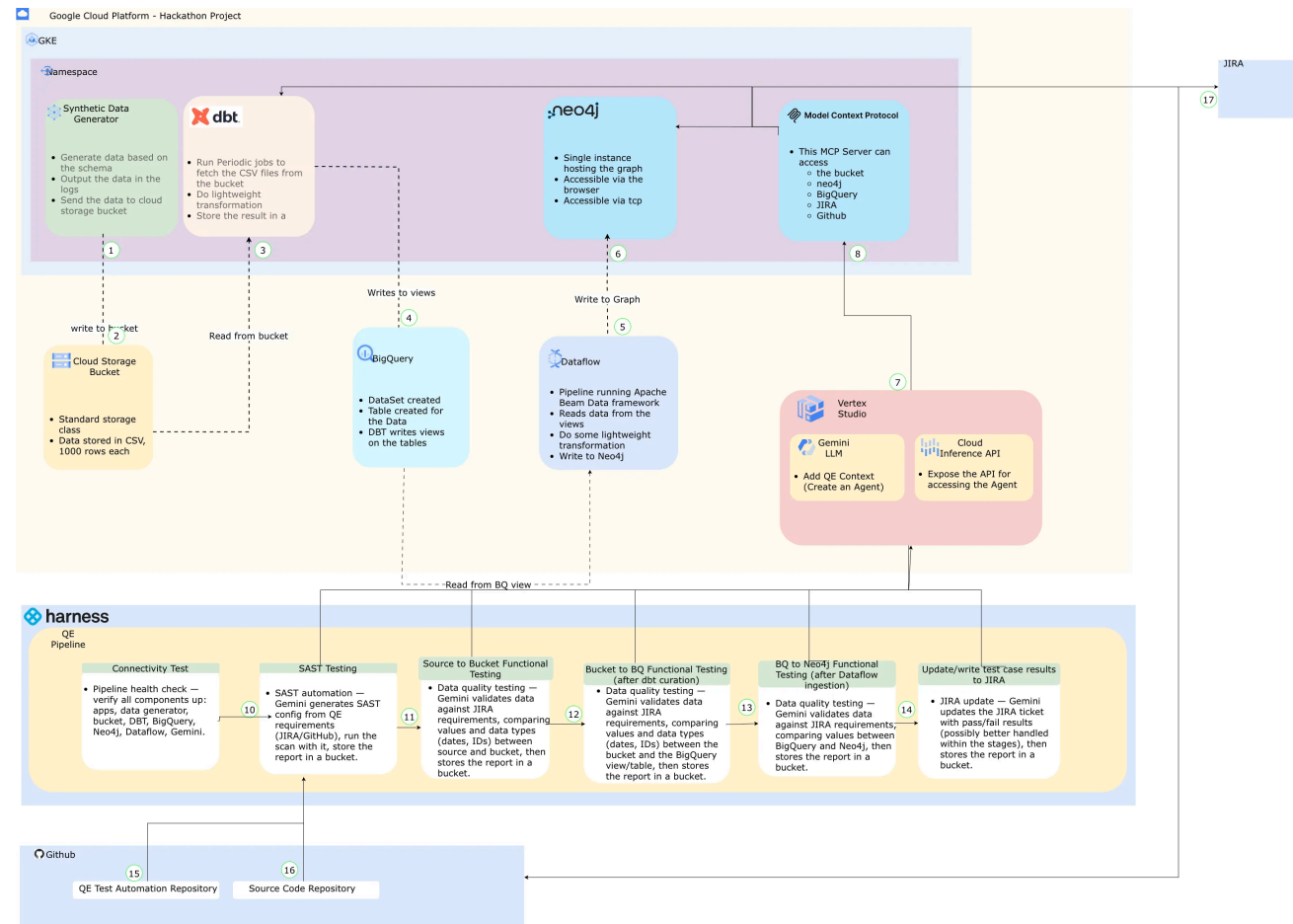


EVIDENCE

 Results on ticket

One agent, three connected worlds.

Helios runs as a single Deployment in the `qe-hack-syndicate` namespace on GKE, brokering Jira, GitHub and Harness while reading the data pipeline it tests (GCS → dbt/BigQuery → Neo4j) and reasoning with Gemini on Vertex.



One codebase: an orchestrator, six sub-agents, eight toolsets.

A quick walk through the repo — the ADK + Gemini orchestrator delegates each quality goal to a specialist sub-agent, every one backed by deterministic toolsets.

qe-agent/agent

```
agent/
├─ root_agent.py      # ADK + Gemini orchestrator
├─ config.py          # models · project · toolset wiring
├─ results.py         # PASS / FAIL / ERROR contract
├─ scenarios/         # 6 deterministic QE suites
│   ├── static_analysis.py
│   ├── coding_standards.py
│   ├── integration.py
│   ├── functional.py
│   ├── non_functional.py
│   └─ runner.py
├─ sub_agents/        # specialist Gemini agents
├─ tools/             # gcs·bq·neo4j·dbt·k8s·jira...
└─ server/jira_webhook.py
```

ORCHESTRATOR → SIX SPECIALIST SUB-AGENTS

static_analysis

coding_standards

integration

functional

non_functional

bdd_authoring

EIGHT DETERMINISTIC TOOLSETS (READ-ONLY)

gcs

bigquery

neo4j

k8s

dbt

jira

harness

github

Live walkthrough — `root_agent.py`, the `sub_agents/` package and the `tools/` MCP layer. Gemini reasons; the toolsets return a strict `PASS / FAIL / ERROR`.



GitHub

Code, generated features and agent-authored pull requests.



Harness

CI/CD pipelines that execute and gate every test suite.



BigQuery

Raw → enriched data the functional & integration suites assert against.



Cloud Storage

Raw CSV landing zone for the synthetic data pipeline.



GKE

From acceptance criteria to evidence — hands-off.

1

A ticket is created

Jira emits `issue_created` to the agent. It verifies the token and reads the acceptance criteria.

```
POST /qe/jira/webhook
```

2

The agent authors the tests

The `bdd_authoring_agent` generates `.feature` files, creates linked Jira *Test* subtasks, and opens a GitHub PR.

3

The ticket moves to "Testing"

An `issue_updated` status transition fires the Harness `bdd_tests` pipeline via a custom webhook trigger.

4

Pipelines run the suites

Harness calls the deterministic `/qe/scenario/...` endpoints and the Cucumber pack — the same code path runs in CI, the agent and the CLI.

5

Evidence is written back

Results are synced onto the ticket's subtasks / Xray execution history. Failing runs are triaged and reconciled into follow-up PRs.

```
POST /qe/jira/{ticket}/sync-results
```

sqlfluff, ruff, bandit, yamllint — zero violations, no hardcoded secrets.



CS1–CS5

Coding standards

Model naming, documented sources, pinned images, resource limits, FK consistency.



I1–I5

Integration testing

GCS↔BigQuery parity, no orphan FKs, Neo4j nodes & relationships, watermark integrity.



F1–F6

Functional testing

Business rules — investment classification, address & phone normalisation, dbt tests.



N1–N2

Non-functional

— LIVE DEMO

Let's watch it run, end to end.

From a single Jira ticket to trusted evidence — the full loop, live against the running PoC.



Create Jira ticket



Agent authors BDD + PR



Move to "Testing"



Harness runs the suites



Evidence back on the ticket

Demo environment · the live PoC in the qe-hack-syndicate namespace on GKE.

Valid product testing

Six deterministic suites assert real business rules against live BigQuery & Neo4j data — strict PASS/FAIL, no false confidence.

2.2



Autonomy

A Jira webhook drives authoring, pipeline execution and result write-back with no human in the loop; failures auto-triage into PRs.

2.3



Easy LBG implementation

Runs only on tooling LBG already operates — Jira/Xray, GitHub, Harness, GKE, BigQuery. A team adopts the template in one sprint.

2.4



User-friendly & User Manual

Plain /qe REST API and a thin qe-cli, backed by a full user manual — README, integration guide, IAM & run-books.

2.5



Whole-team understanding

Deterministic design and documented runbooks mean every one of the four squad members can explain, run and extend the agent.

2.6



Maintainability & scalability

Single shared codebase, pinned images, Workload Identity, config-driven per team. Built once; scales to ~40 teams at marginal cost.

2.7



Open source support



Senior DevOps Engineer

Platform & delivery

- GKE, Workload Identity, RBAC
- Harness pipelines & triggers
- Secrets, networking, observability



Backend Engineer

Python developer

- ADK agent & toolsets
- Deterministic /qe API
- Jira / GitHub / Harness integration



Senior Quality Engineer

Test strategy & BDD

- Scenario design & coverage
- Cucumber / Xray integration
- Quality gates & evidence model



A few focused hours a week — not full-time headcount.

ROLE	LOADED £/HR	5–6 HRS/WK
Senior DevOps Engineer	£55	£275–£330
Backend (Python) Engineer	£48	£240–£288
Senior Quality Engineer	£52	£260–£312
Customer Journey Manager	£50	£250–£300
Full squad / week	£205/hr	£1,025–£1,230

Fully-loaded hourly = base salary + ~30% employer NI, pension & overheads, over ~1,800 productive hours/yr. UK 2025/26 perm rates, financial-services / fintech market. Each engineer commits just 5–6 focused hours per week.

~£13k–£16k

per quarter (~13 weeks)

Whole squad, 5–6 hrs each per week

~£27k–£32k

2-quarter MVP (end-to-end)

Production-ready, full squad

A fraction of a single FTE

At 5–6 hours each per week, the all-in people cost lands around **£27k–£32k** for a fully delivered, production-ready MVP — well under one full-time engineer for a quarter, and a one-off build that every team then reuses.

GKE.

SERVICE	TIER	~ / MONTH
GKE node pool (2× e2-standard-2)	GCP	£80
Load balancers + networking (2 LBs, egress)	GCP	£35
Vertex AI — Gemini 2.5 Flash (agent calls)	GCP	£20
BigQuery + GCS + Artifact Registry	GCP	£10
Persistent disks (Neo4j + pipeline)	GCP	£5
Harness	Free trial	£0
Jira / Xray	Free tier	£0
GitHub	Free	£0
Neo4j (community) · dbt Core	Free / OSS	£0
Total PoC run cost		≈ £150 / month

≈ £150 / month

All-in cloud spend for the live PoC

≈ £40–£50 for a typical Hackathon run window. Self-hosted Neo4j on GKE keeps graph costs at zero beyond compute.

Model choice for the PoC

In LBG the platform licences (Harness, Jira/Xray, GitHub Enterprise, GCP org) are already paid — so the marginal cost is a hardened, HA agent footprint shared across many teams. Sizing assumes **~200 pipeline runs/day across ECP** (~4,400 / month).

COMPONENT	NOTE	~ / MONTH
GKE (regional, HA agent + supporting)	Shared cluster, marginal	£500
Neo4j (AuraDB Pro / HA cluster)	Production graph	£400
Vertex AI — Gemini (authoring + triage)	~4,400 runs/mo, ~30% LLM-assisted	£200
BigQuery (scaled data + queries)	Storage + on-demand	£200
Networking, LBs & egress	HA ingress	£120
Logging, monitoring & secrets	Observability	£80
Harness · Jira/Xray · GitHub Enterprise	Existing LBG licences	£0*
Marginal platform cost		≈ £1,500 / month

< £1.6k

per month, whole platform
Shared across all adopting teams

~£40

marginal cost per team / month
At ~40 teams on the shared template

Model strategy & build-once economics

The PoC runs on **Gemini 2.5 Flash**: for the MVP we step up to **the latest Gemini Pro** model for deeper reasoning at scale. Gemini is invoked only for authoring and failure triage, so each extra

From Hackathon PoC to reusable platform.

Q1 · WEEKS 1–4

Harden the core

Productionise the agent & toolsets, secrets, RBAC, CI/CD on the LBG estate.

Q1 · WEEKS 5–8

Prove on a real service

Wire one ECP team's Jira project end-to-end; validate evidence & gating.

Q1 · WEEKS 9–12

Template & document

Package the reusable template, onboarding guide and Xray integration.

Q2 · SCALE

Roll out & measure

Onboard further teams, track value, and prepare the Group-wide case.

1 sprint

for a team to adopt the
template

1–2 Q

to a production-ready MVP

~40 teams

addressable on the shared
pattern

— WHY IT LANDS AT LBG

Faster feedback, consistent quality, trusted evidence — on tools we already run.

A lean squad delivers a reusable MVP in 1–2 quarters for around **£27k–£32k** of people cost — just 5–6 hours each per week — running for **under £1.6k a month** across the whole platform — roughly **£40 per team**. Built once, adopted in a sprint, and ready to scale Group-wide.

● Reduce duplicated effort

● Faster, consistent coverage

● Auditable evidence

● Group-wide foundation